

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN SOILS

MODULE - 3

SOIL TYPES

SOIL PROFILE

SOIL TYPES

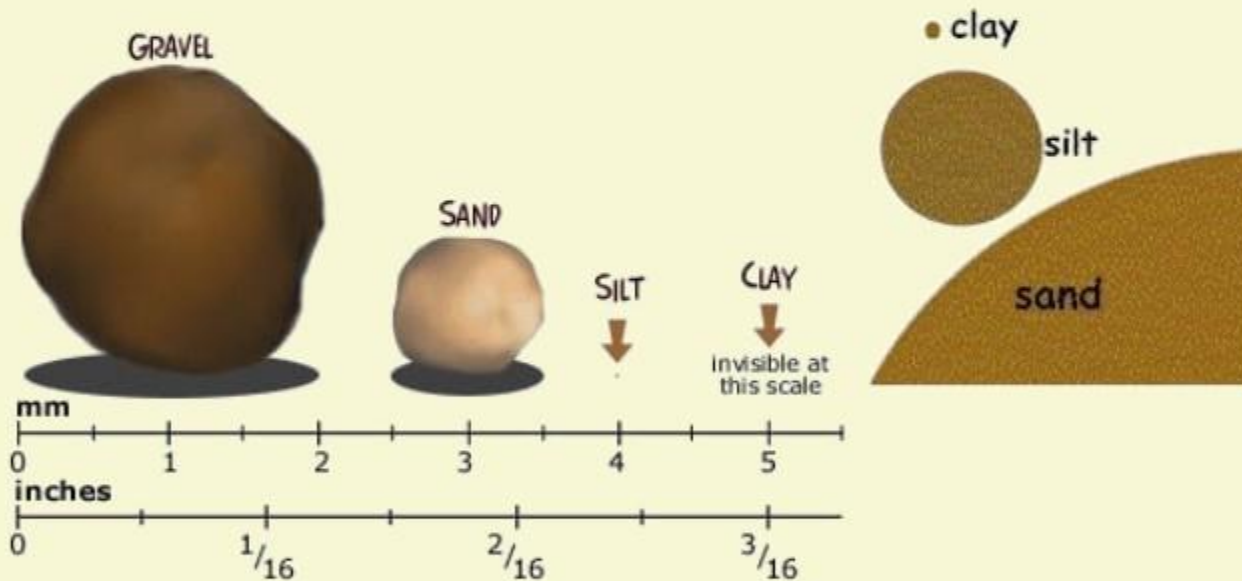
- 1 The soil is classified is on the basis of proportions of particles of various sizes.
 - If soil contains greater proportion of big particles, it is called sandy soil.
 - If the proportion of fine particles are relatively higher, then it is called clayey soil.
 - If the amount of large and fine particles is about the same, then the soil is called Loamy soil.

- 2 Water can drain quickly through the spaces between the sand particles. So sandy soil tend to be light, well aerated and dry.

- 3 Clay particles, being much smaller, pack tightly together leaving little space for air. Unlike sandy soil water can be held in the tight gaps between the particles of clay.

- 4 So clay soils have little air but they are heavy as they hold more water than the sandy soils.

SOIL TYPES



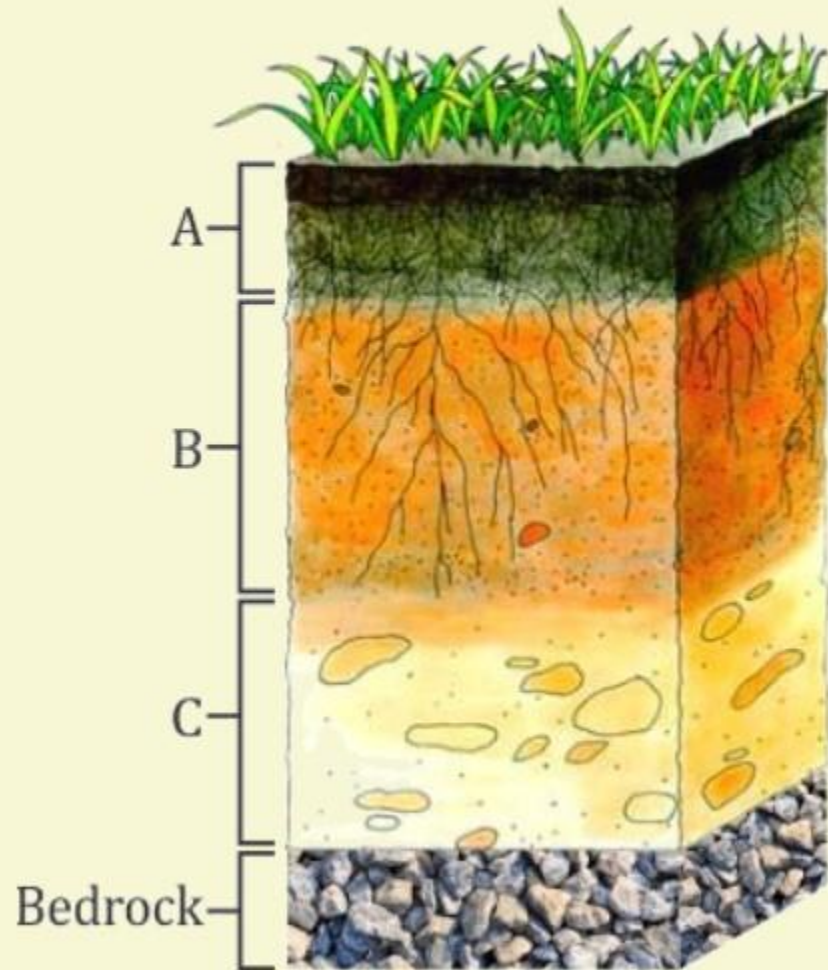
5 The best top soil for growing plants is **Loamy soil**, which is a mixture of sand, clay and another type of particle known as Silt.

6 Silt occurs as a deposit in river beds. The size of the silt particles is between those of sand and clay.

7 The loamy soil also has humus in it and it has the right water holding capacity for the growth of plants.

SOIL TYPES

SOIL PROFILE



1 A vertical cross-section cutting through different layers of the soil is called the soil profile.

2 Each layer differs in feel (texture), colour, depth and chemical composition.

3 A Soil Horizon is generally parallel to the soil surface, whose physical characteristics are different from layers above and beneath.

4 Horizons are defined in most cases by obvious physical features like colour and texture.

SOIL TYPES

SOIL PROFILE

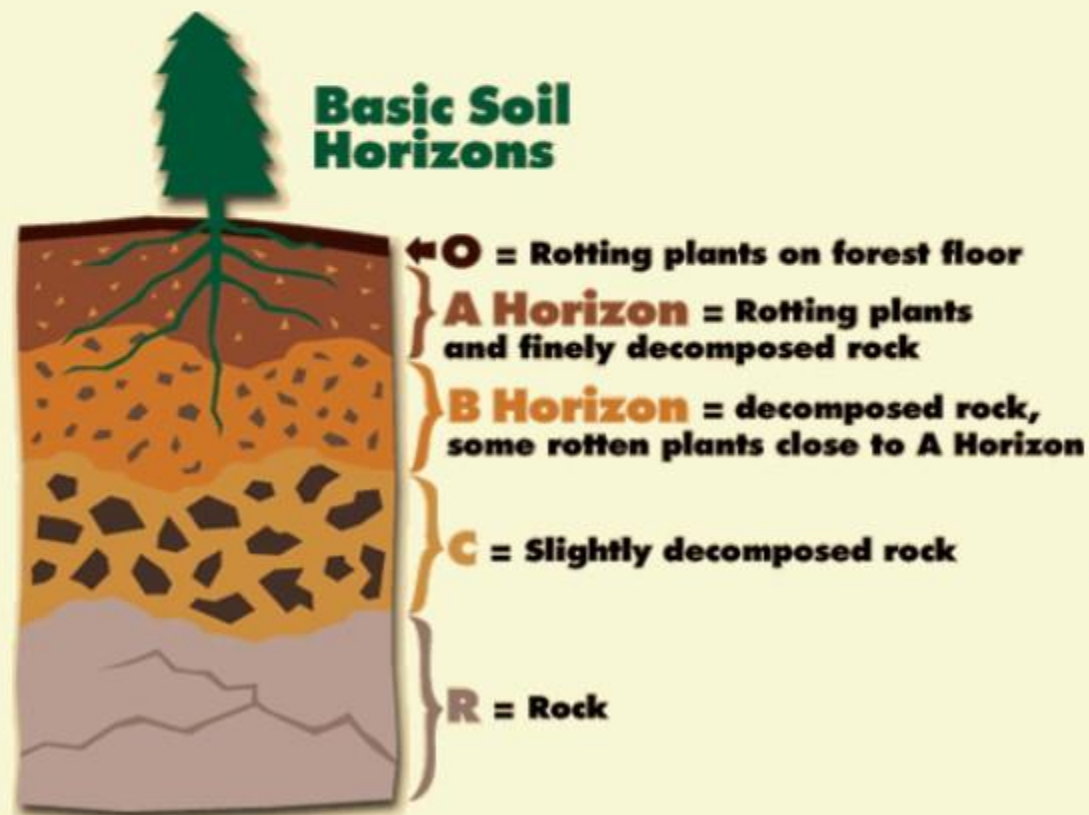
- 5 The A-horizon is dark in colour because it is rich in humus and minerals.

- 5 The humus makes soil fertile and provides nutrients to growing plants.

- 6 This layer of top soil is generally soft, porous and can retain more water. It is known as the **A-Horizon**.

- 7 The next layer has a lesser amount of humus but more amount of minerals. This layer is generally harder and more compact and is called the **B-horizon** or the middle layer.

- 8 The third layer is the **C-horizon**, which is made up of small lumps of rocks with cracks.





SOIL TYPES

SOIL PROFILE

1

O-HORIZON



1 This layer dominated by organic material.

2 Some O-layers consist of **undecomposed or partially decomposed litter** such as leaves, twigs, mosses and lichens.

3 They may be on top of either mineral or organic soils.

SOIL TYPES

SOIL PROFILE

2 A-HORIZON OR SURFACE SOIL

- 1 It is part of top soil, in this layer **organic matter is mixed with mineral matter.**

- 2 It is the layer of mineral soil with the most organic matter accumulation and soil life.

- 3 At the base, this layer gets depleted of iron, clay, aluminum, organic compounds and other soluble constituents.

- 4 When **depletion is pronounced a light colored E sub-surface soil horizon** is present at the base of A-Horizon.



● (humus or organic)

Decomposing leaves and lots of organic matter

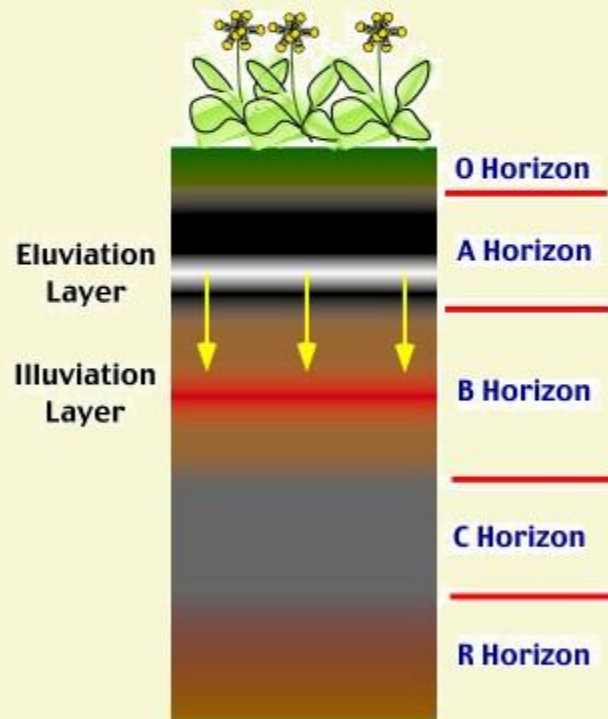
A (topsoil)

Contains lots of roots, and minerals for growing plants

SOIL TYPES

SOIL PROFILE

3 E-HORIZON OR ELLUVIATED LAYER



1 The letter **E** stands for **Eluviated layer**. It is the original layer that has been significantly leached off clay, iron and aluminum oxides which leaves a concentration of resistant minerals.

2 These are present in older, well developed soils and generally occur between the layers of A & B-Horizons.

SOIL TYPES

SOIL PROFILE

4 B-HORIZON OR SUBSOIL



B (subsoil)

Minerals from upper horizons stop here

C (parent material)

Earth's surface that soils developed from

1

It is the subsurface layer reflecting chemical or physical alteration of parent material.

2

This layer accumulates all the leached materials from A & E horizons. Thus iron, clay aluminum and other organic compounds accumulate in this horizon.

3

Thus this horizon is known as the zone of Illuviation opposite of Eluviation.

SOIL TYPES

SOIL PROFILE

5

C-HORIZON



B (subsoil)

Minerals from upper horizons stop here

C (parent material)

Earth's surface that soils developed from

1

Weathered parent material accumulate in this layer of C-Horizon.

2

It is a layer of large broken rocks.

3

This layer may accumulate the more soluble compounds and inorganic material.

SOIL TYPES

SOIL PROFILE

6 R-HORIZON OR BED ROCK



C (parent material)

Earth's surface that soils developed from

R (bedrock)

1 This layer denotes the layer of **consolidated bed rock**, at the base of soil profile. Unlike the above layers, R-Horizon largely comprises of **continuous masses of hard rock**.

2 Soil formed in situ will exhibit strong similarities to this bed rock layer.

3 These areas of bed rock are under 50 feet from other profiles.

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